21 There are 12 counters in a bag.
There is an equal number of red counters, blue counters and yellow counters in the bag.
There are no other counters in the bag.

3 counters are taken at random from the bag.

(a) Work out the probability of taking 3 red counters.

There are 12 ÷ 3 = 4 red counters. No replacement.

$$P(\text{three red}) = \frac{4}{12} \times \frac{3}{11} \times \frac{2}{10} = \frac{1}{55}$$

1 mark
Multiply the three changing probabilities

1 mark
Correct product

$\frac{1}{55}$ **(2)**

The 3 counters are put back into the bag.

Some more counters are now put into the bag.
There is still an equal number of red counters, blue counters and yellow counters in the bag.
There are no counters of any other colour in the bag.

3 counters are taken at random from the bag.

(b) Is it now less likely or equally likely or more likely that the 3 counters will be red?
You must show how you get your answer.

For example, add 1 of each colour: 5 red out of 15.

$$P(\text{three red}) = \frac{5}{15} \times \frac{4}{14} \times \frac{3}{13} = \frac{2}{91}$$

1 mark
Calculate a relevant probability after adding counters

2/91 = 0.021978... > 1/55 = 0.018181...

More likely. The first red chance stays 1/3; with more of each colour, the later red chances increase.

1 mark
Compare the probabilities and give the conclusion

(2)

(Total for Question 21 is 4 marks)

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